

Title:

Tracing the Thermal Footprint of Urban-Industrial Expansion: A Satellite-Based Analysis of Surface Urban Heat Island Dynamics in Savar, Bangladesh

Aim:

To investigate how long-term urban expansion influences land surface temperature dynamics and surface urban heat island (SUHI) intensity in Savar, Dhaka, and to assess whether observed SUHI changes reflect linear urban growth, non-linear surface thermal amplification, or atmospheric background variability, using multi-temporal Landsat observations and percentile-based thermal metrics.

RQ1: How has urban land cover expanded in Savar (2000–2025) and how has this changed the balance of built-up, vegetation, and water surfaces?

RQ2: How have LST patterns changed across time, especially between urban and rural vegetation surfaces?

RQ3: How have SUHI intensity, spatial extremes, and hotspot persistence evolved over time, and do these changes indicate linear or nonlinear amplification behavior?

Atmospheric control vs surface-driven amplification (ERA5 validation):

To what extent are SUHI variations driven by background atmospheric conditions vs urban surface amplification?

3. Data and Methodology

3.1 Study Area

The study focuses on Savar Upazila located on the north-western peri-urban fringe of Dhaka District, Bangladesh, functioning as a transitional zone between the Dhaka metropolitan core and the adjacent rural floodplain landscapes. The spatial coordinates of the study area lie between 23.74°-24.04° N latitude and 90.19°-90.36° E longitude (WGS 84). Savar is characterized by mostly low-lying landscape combined with rivers, wetlands, agricultural fields, industrial estates, and rapidly expanding residential zones. Climatically, Savar experiences a tropical monsoon regime characterized by a hot pre-monsoon season (March-May), monsoon rainfall (June-September), and a relatively cooler dry season (November-February), (Alam et al., 2023).

3.2 Data Sources

In this study, multi-temporal Landsat imagery was obtained from the United States Geological Survey (USGS) EarthExplorer platform. The analysis covers six temporal blocks spanning 2000 to 2025. Each block represents a two-year composite of warm-season imagery (April-July). A two-year compositing strategy was adopted due to the limitation of cloud-free scenes and to minimize the influence of cloud contamination, data gaps, and scene-level noise.

2000–2001 (Landsat 7 ETM+)

2005–2006 (Landsat 5 TM)

2010–2011 (Landsat 5 TM)

2015–2016 (Landsat 8 OLI/TIRS)

2020–2021 (Landsat 8 OLI/TIRS)

2024–2025 (Landsat 9 OLI-2/TIRS-2)

All Landsat datasets used in this study were atmospherically corrected surface reflectance products (Collection 2 Level-2). The thermal infrared bands were used to derive land surface temperature (LST). In addition, ERA5-Land 2 m air temperature (t2m) data were incorporated to provide background atmospheric context and to examine whether observed SUHI variations were influenced by broader meteorological conditions.

3.3 Pre-Processing and Compositing

All image processing and analysis were conducted using Google Earth Engine (GEE). Pre-processing steps included cloud masking based on QA bands, seasonal filtering of April-July scenes, and percentile-based compositing. As we found frequent cloud contamination and data gaps during the warm season, a percentile approach was adopted instead of simple mean compositing. Therefore, 75th percentile (p75) was selected to better represent elevated surface temperature conditions while reducing the influence of residual noise and extreme outliers.

Compared to mean or median composites, p75 allows our analysis improved tracing of higher-end surface temperature behavior, which is particularly relevant for SUHI analysis.

The p75 percentile was calculated as:

$$p_q(X) = \inf \{x \in \mathbb{R}: F_X(x) \geq q\}$$

where $q = 0.75$ for p75 and $q = 0.90$ for p90, and F_X represents the empirical cumulative distribution function (CDF) of pixel-wise values within each temporal block.

Moreover, land surface temperature (LST) and spectral indices were derived from atmospherically corrected surface reflectance products. LST was calculated from Landsat thermal infrared bands using standard brightness temperature (BT) conversion and radiative transfer procedures, incorporating sensor-specific calibration constants.

3.4 Spectral Indices and Land Surface Classification

Instead using true land use land cover data we classified dominant surface types, using following indices and which were calculated:

$$\begin{aligned} \text{NDVI} &= \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}} \\ \text{NDBI} &= \frac{\text{SWIR} - \text{NIR}}{\text{SWIR} + \text{NIR}} \\ \text{MNDWI} &= \frac{\text{Green} - \text{SWIR}}{\text{Green} + \text{SWIR}} \end{aligned}$$

Surface dominance was defined using index-based thresholds and remain constant for all year blocks. We used threshold limits for Urban-dominant: $\text{NDBI} > -0.05$ and $\text{NDVI} < 0.25$, Vegetation-dominant: $\text{NDVI} > 0.20$, and Water-dominant: $\text{MNDWI} > 0$ and $\text{NDVI} < 0.25$ and $\text{LST} < 35^\circ\text{C}$. However, water surfaces were excluded from SUHI calculations. To accurately identify persistent water bodies and minimize misclassification, the JRC Global Surface Water dataset was used to delineate long-term water-dominant areas.

3.5 Surface Urban Heat Island (SUHI) Intensity

In this study, SUHI intensity was calculated as:

$$\text{SUHI} = \text{LST}_{\text{urban}} - \text{LST}_{\text{vegetation}}$$

Where, urban and vegetation reference pixels were defined based on dominant index classifications. Mean SUHI (p75) was used to represent intensity and extreme SUHI (p90) was

used to identify hotspot thresholds, though, totspots were defined as the upper 10% ($\geq p90$) of SUHI values for each year block.

3.6 Hotspot Persistence and Spatial Density

Hotspot persistence was assessed by counting the number of periods in which a pixel exceeded the p90 SUHI threshold (range: 0-6). To evaluate spatial consolidation, a 500 m moving window density analysis was conducted to identify clusters of recurrent hotspots over the entire study area. However, persistence classes were categorized as:

Low (repetition at least 1-2 periods)

Moderate (repetition at least 3 periods)

High (repetition at least 4-5 periods)

Very high (≥ 5 periods)

3.7 Atmospheric Contextualization

Finally, ERA5-Land 2 m air temperature (t2m) data were extracted for corresponding April-July periods. We used ERA5 data to assess whether observed SUHI trends were attributable to background atmospheric warming or surface-driven amplification processes instead explicitly validating the whole SUHI study.

4. Results

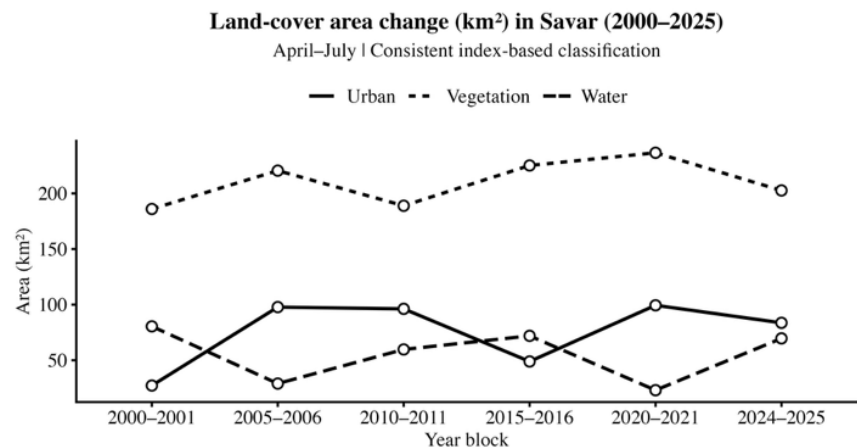
Index-based land-cover dynamics and urban dominance in Savar:

Figure 1 and Table 2 present an overview of the index-based land-cover dynamics in Savar Upazila across six temporal blocks (April–July) from 2000–2001 to 2024–2025. Dominant surface characteristics were derived using consistent spectral indices that represents urban, vegetation, and water signatures. The results indicate a clear long-term reorganization of surface dominance, marked by episodic intensification of urban signatures, substantial variability specially in water-dominant surfaces, and persistent but dynamic vegetation dominance.

Year block	Urban (km ²)	Urban (%)	Vegetation (km ²)	Vegetation (%)	Water (km ²)	Water (%)
2000–2001	27.25	7.83	186	53.42	80.48	23.11
2005–2006	97.79	28.08	220.47	63.32	29.05	8.34
2010–2011	96.22	27.63	188.9	54.25	59.72	17.15
2015–2016	48.9	14.04	225.12	64.65	71.93	20.66
2020–2021	99.41	28.55	236.52	67.93	23.09	6.63
2024–2025	83.64	24.02	202.63	58.19	69.65	20

Urban-dominant surfaces exhibit an overall upward tendency through time. In 2000–2001, urban-dominant signatures cover approximately 28 km² (about 8% of the study area), increasing to nearly 85 km² (around 23 to 24%) by 2024–2025). This temporal trajectory is distinctly non-

monotonic, with two phases of intensified urban dominance: an early expansion occurred in between 2000–2001 and 2005–2006, and a second peak around 2020–2021, when urban-dominant surfaces reach their maximum extent (≈ 100 km²). A temporary contraction is noticed in 2015–2016



which interrupts this trend, indicating short-term moderation in urban surface expression rather than a reversal of long-term change.

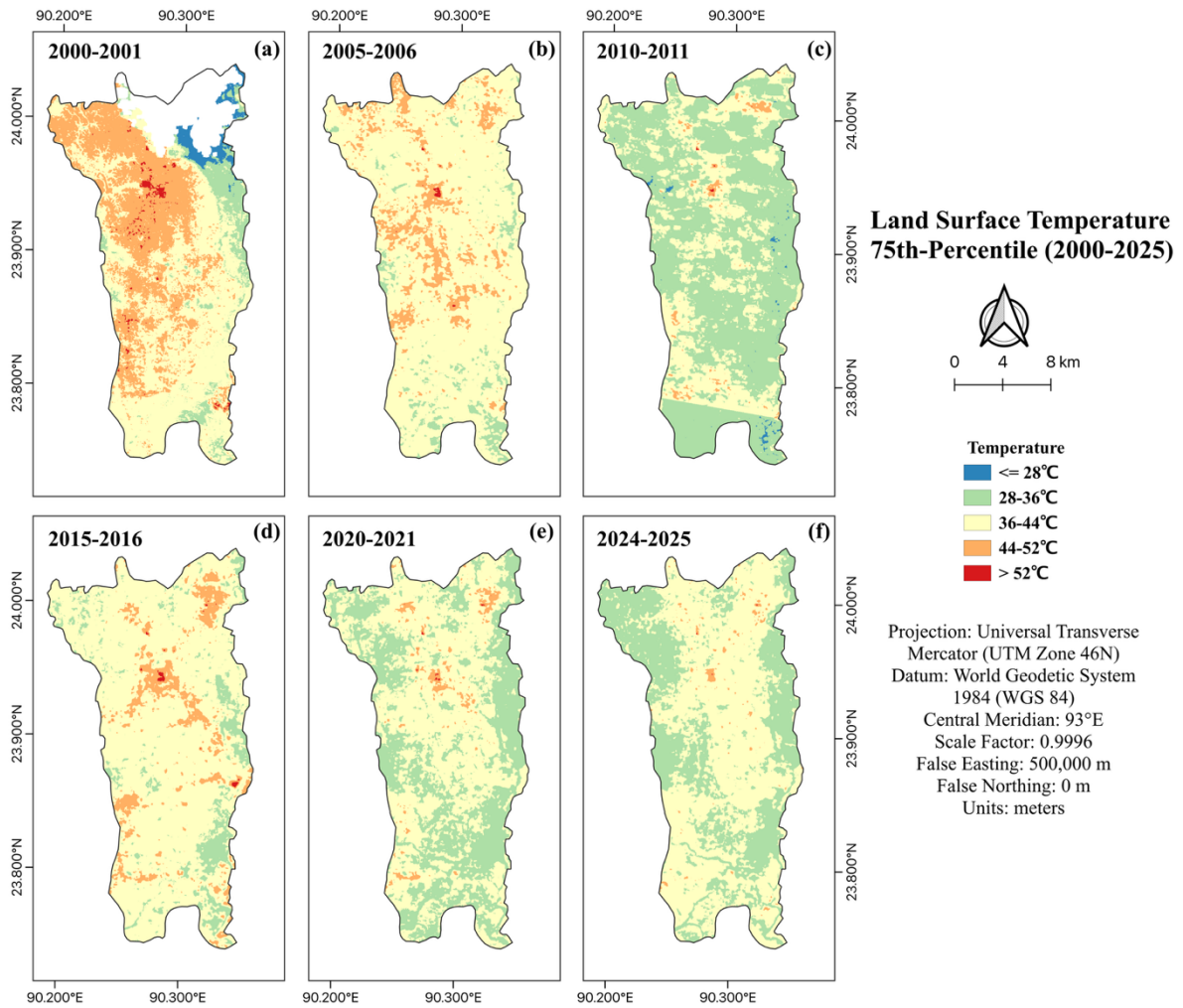
On the other hand, vegetation-dominant surfaces remain the most extensive class throughout the study period but present a pronounced inter-block variability. Periods of strong urban surface expression (2005-2006 and 2020-2021) coincide with elevated vegetation-dominant extent (≈ 220 and 237 km^2), that indicates the increases in urban-dominant signatures do not uniformly correspond to reductions in vegetation-dominant surface signals. Instead, these periods are primarily characterized by contractions in water-dominant surfaces ($\approx 29 \text{ km}^2$ in 2005-2006 and $\approx 23 \text{ km}^2$ in 2020-2021), suggesting a redistribution among dominant surface signatures rather than direct urban-vegetation displacement. Following the 2020-2021 peak, vegetation dominance weakens in 2024-2025 ($\approx 203 \text{ km}^2$), which eventually highlights the non-linear nature of index-based surface characterization.

Water-dominant surfaces show pronounced temporal variability and a consistent inverse association with urban surface dominance. Periods of intensified urban dominance correspond to the clear reductions in water-dominant extent, whereas intervals of reduced or stabilized urban dominance (2010-2011 and 2015-2016) coincide with recovery of water-dominant surfaces, exceeding 70 km^2 in some blocks. This pattern indicates that temporal reconfiguration of surface dominance in Savar is facilitated primarily through shifts in water-related spectral signatures rather than uniform displacement of vegetation-dominant surfaces.

Temporal Changes in LST and Urban-Vegetation Contrast (April-July, p75):

The p75 LST maps reveal obvious inter-block changes in the dominant temperature classes (Figure 5). Blocks 2000-2001, 2005-2006, and 2015-2016 (Figures 5a, b, d) are characterized by widespread dominance of the $36\text{-}44^\circ\text{C}$ class, whereas noticeable localized areas reaching higher temperature ranges. Most of the areas are fall in under the class $36\text{-}44^\circ\text{C}$ and their common hot zone in central Savar (e.g., Savar city area, Bipail), Ashulia industrial zones and Amin Bazar nearby Dhaka at southeastern part of the Savar upazila.

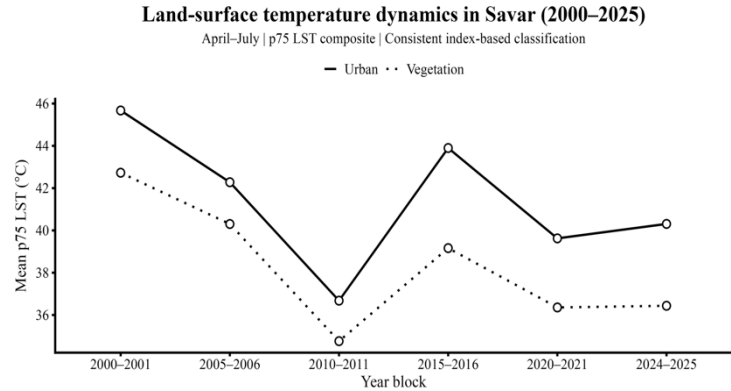
However, block 2000-2001 shows an extensive heat area at the western and northwestern part of the study area. In contrast, 2010-2011 (Figure 5c) shows a clear shift toward cooler classes ($28\text{-}36^\circ\text{C}$ and $36\text{-}44^\circ\text{C}$), where high-temperature categories have become spatially limited. The later blocks (2020-2021 and 2024-2025; Figures 5e, f) exhibit stable but more spatially fragmented warm signatures, which indicate the persistence of elevated p75 LST in localized areas rather than across the broader landscape, which eventually does not satisfy expectation of urban growth insight. However, the distinct urban, climatic, and geographic conditions establish Savar Upazila as a climatically sensitive peri-urban system, where land-cover reconfiguration including spectral variations can rapidly amplify surface thermal extremes.



Both urban and vegetation-dominant surfaces demonstrate their highest mean p75 LST values in 2000-2001, followed by a pronounced decline toward 2010-2011, again a secondary increase in 2015-2016, and comparatively lower but also stable values in 2020-2021 and 2024-2025 (Figure 6; Table 7). This temporal evolution is broadly similar for both surface classes i.e., urban and rural like, despite persistent differences in absolute

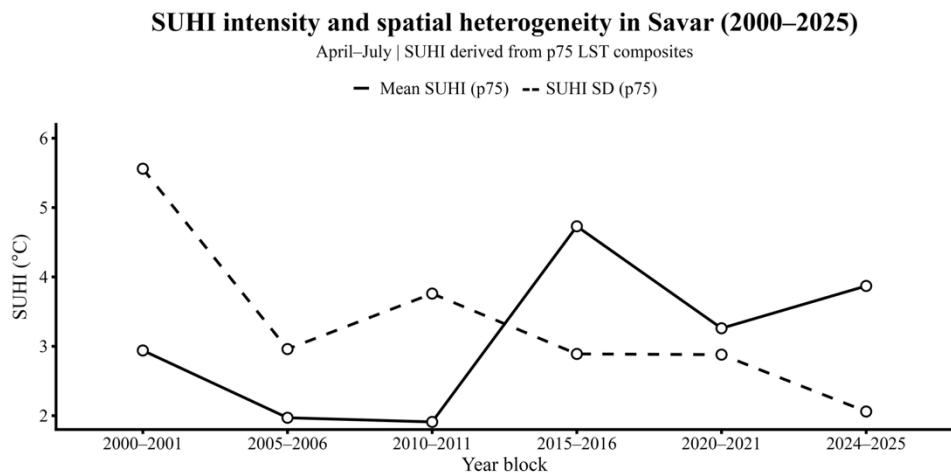
	UrbanMean	RuralVegMean	Delta
Year block	LST_p75_C	LST_p75_C	LST_C
2000-2001	45.67	42.73	2.94
2005-2006	42.28	40.30	1.98
2010-2011	36.68	34.76	1.91
2015-2016	43.90	39.16	4.73
2020-2021	39.62	36.36	3.27
2024-2025	40.31	36.44	3.87

temperature levels, indicating that periods of elevated or suppressed p75 LST affect urban and vegetation-dominant signatures concurrently. The coexistence of high p75 LST in both classes during the early period, followed by simultaneous cooling toward 2010-2011, produces a non-monotonic thermal trajectory that does not directly mirror the progressive urban expansion which was documented in the [first section](#). In contrast to this parallel movement in absolute p75 LST, the urban-vegetation thermal contrast (Δ LST) displays a distinct temporal pattern ([Figure 7; Table 7](#)). Δ LST remains relatively small in the earlier and intermediate blocks (2005–2006 and 2010–2011, $\approx 2^\circ\text{C}$) but increases sharply in 2015–2016 ($\approx 4.7^\circ\text{C}$) and remains elevated in the most recent periods (2020–2021 and 2024–2025, $> 3^\circ\text{C}$), even as absolute p75 LST levels stabilize or decline.



SUHI Intensity, Extremes, and Hotspot Persistence (April–July, p75):

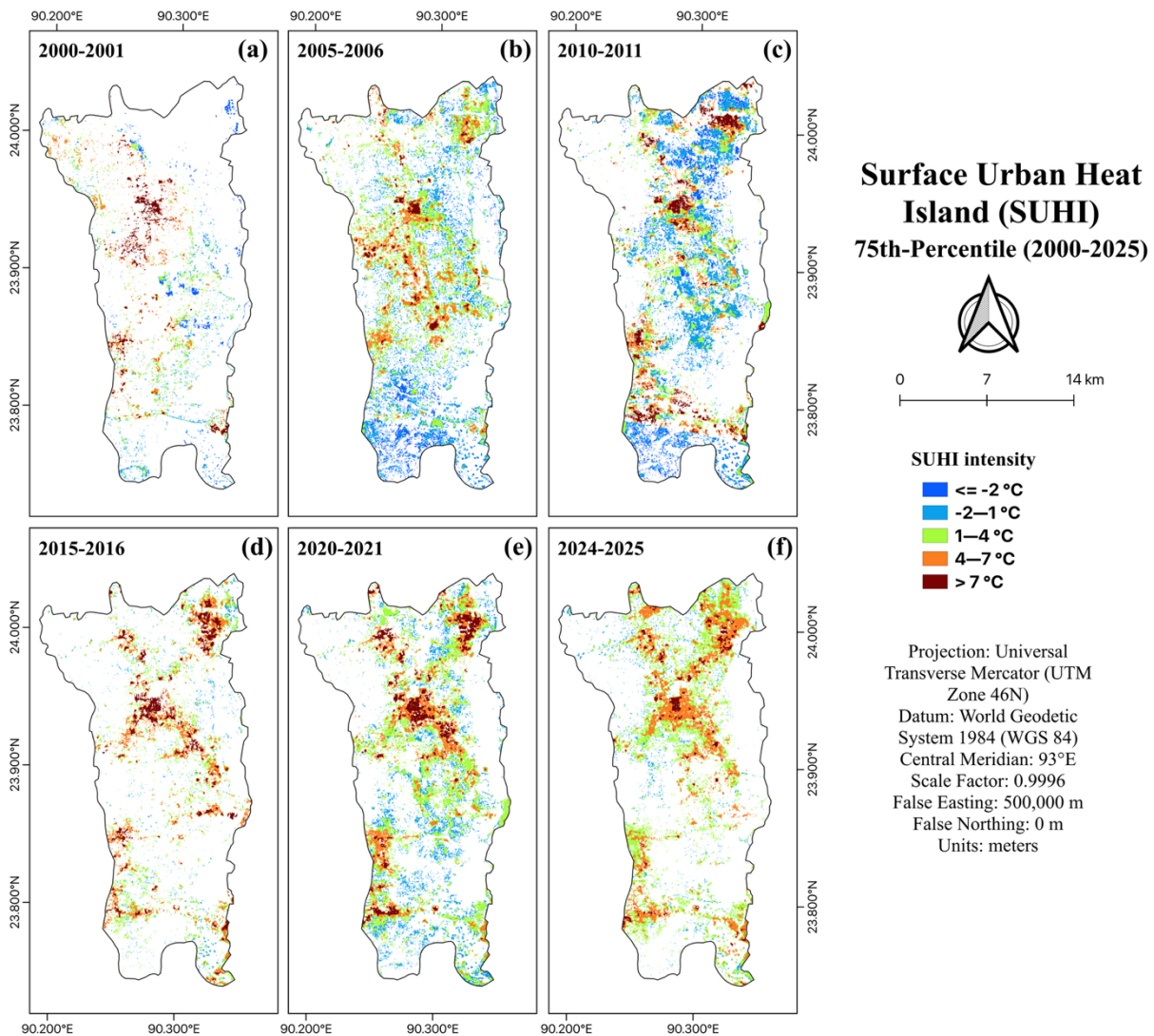
Our findings of SUHI behavior in Savar shows episodic amplification rather than long term steady intensification with a clear separation between changes in the central tendency and the upper tail. Mean SUHI derived from p75 LST



declines from 2.94°C (2000–2001) to ~ 1.9 to 2.0°C (2005–2011), before rising sharply to a maximum of 4.73°C (2015–2016) and remaining elevated through the recent blocks, i.e., 2020–2021 (3.26°C) and 2024–2025 (3.87°C) ([Fig. 8; Table RQ3](#)). In parallel, spatial heterogeneity behaves in the opposite direction; ([Fig. 8; Table](#) show SUHI SD is highest in 2000–2001 (5.56°C) and progressively compresses to $\sim 2.9^\circ\text{C}$ (2015–2021) and 2.06°C (2024–2025). This reveals a clear contrast in SUHI behavior where recent periods show higher average SUHI while spatial differences become smaller.

The block-wise SUHI maps reveal a systematic reorganization of the spatial structure of urban heat across Savar ([Fig. 11](#)). In the earlier blocks, high SUHI values are largely restricted to

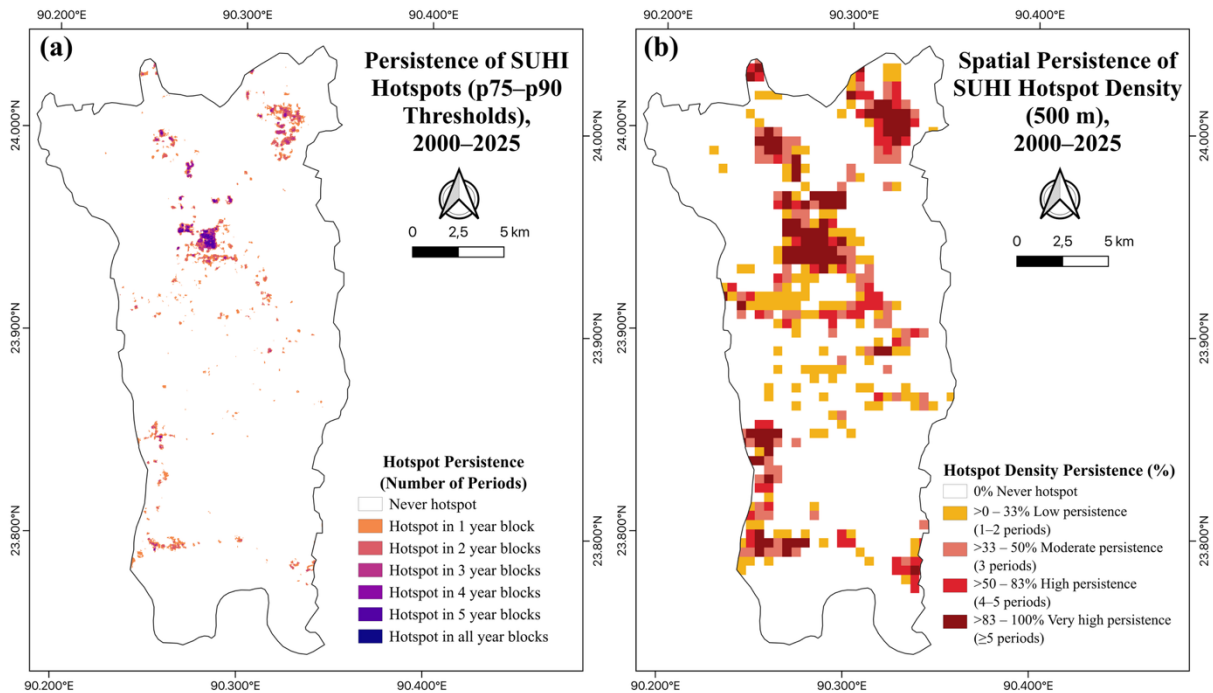
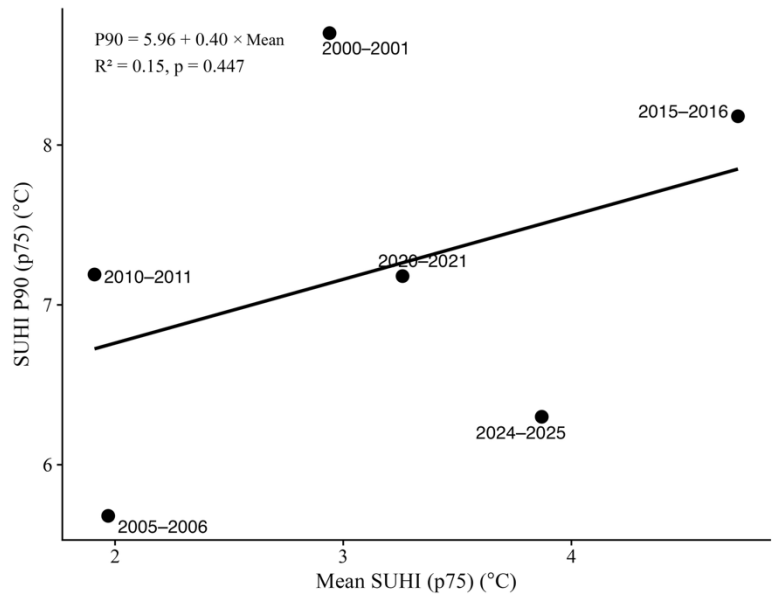
a limited number of discrete locations, e.g., dense industrial zones, high impervious areas, embedded within extensive areas of weak to moderate SUHI, and these early blocks reflect a highly heterogeneous heat pattern. In contrast, later blocks are characterized by increasingly coherent and spatially contiguous high-intensity SUHI zones, most prominently along the central development corridor and within the north-eastern sector, that indicates a transition from localized to more widespread urban heat expression. However, from this progression an important deviation is also evident in 2010–2011, where extensive weak or negative SUHI conditions coexist with a small number of pronounced hotspots.



The extreme SUHI values show that average urban heat and the strongest heat hotspots do not always change together. In some years, the overall SUHI is relatively low, but very hot local areas still exist. For example, in 2010–2011, the average SUHI is one of the lowest in the study

period (1.91 °C), yet the extreme SUHI value (p90 = 7.19 °C) remains high (Table RQ3). This means that even when the city is not very hot overall, a few places can still experience very intense heat. This pattern shows that extreme SUHI does not increase in a smooth or proportional way as average SUHI changes. This nonlinearity is further supported by the weak scaling between mean SUHI and p90 SUHI (Fig. 9). The regression relationship ($P90 = 5.96 + 0.40 \times \text{Mean}$) explains little variance ($R^2 = 0.15$; $p = 0.447$), which denotes that extreme SUHI does not increase in a simple linear proportion to mean SUHI.

Scaling Between Mean and Extreme SUHI (P90) Intensity Across Time
 Regression-based assessment of linear versus nonlinear SUHI amplification



We also have conducted hotspot persistence analysis, and which indicates that these extremes are spatially anchored rather than randomly distributed (Fig. 10). The pixel-level persistence map shows that most hotspot pixels occur in only one or two blocks, but a smaller set of multi-block cores recur repeatedly, especially within the central zone, north-eastern cluster and southeastern cluster (Fig. 10a). The 500 m hotspot-density persistence surface further strengthens this inference through identifying a central north-south belt with moderate-to-very high persistence and additional persistent nodes toward the margins (Fig. 10b).

5. Discussion

Urban Expansion and Surface-Dominance Reconfiguration:

Land-cover dynamics in Savar between 2000 and 2025 exhibit a non-linear, mosaic-like reconfiguration of urban, vegetation, and water-dominant surfaces that departs from classical models of monotonic urban growth. Although urban-dominant surfaces increased substantially, this expansion did not consistently coincide with vegetation loss. Vegetation dominance remained high or increased during major urban growth phases (e.g., 2005-2006 and 2020-2021), contradicting a simple urban-vegetation substitution model.

This apparent contrast reflects both the peri-urban character of Savar and the dominance-based classification framework. In the Dhaka fringe, built-up expansion commonly coexists with bare land, cropland, homestead vegetation, and tree cover, producing a mixed-use mosaic in which vegetation can remain spectrally dominant despite ongoing urbanization (Ullah et al., 2019); (Rahman et al., 2022). The April-July observation window further coincides with early monsoon greening, i.e., as a tropical monsoon-controlled climate region, the area experiences abundant growth of lush green vegetation during the early monsoon season (Awasthi et al., 2023). Which enhances NDVI dominance even within developing urban areas. Consequently, increases in vegetation dominance indicate surface greenness amplification rather than the absence of urban encroachment. However, the findings of inverse fluctuations between urban and water dominance are primarily driven by seasonal hydrological variability and the dominance-based spectral framework, rather than repeated physical infilling or restoration. Though the water dominant surface trajectory shows comparatively contrasting with strong fluctuations. This pattern may include wetland infilling in Savar, Gazipur, and eastern Dhaka, where low-lying lands are preferentially converted for residential and industrial development (Ullah et al., 2024); (Sultana, 2009).

Inter-annual differences in surface moisture during the pre-monsoon to early monsoon period alter NDVI, NDWI, and NDBI contrasts, causing transitional pixels to shift dominance. The temporary decline in urban dominance in 2015-2016 therefore reflects climate-sensitive suppression of urban spectral signals under wetter conditions, with concurrent increases in vegetation and water dominance which are linked to monsoon-driven regrowth, surface saturation, seasonal inundation, and transitional construction stages (Kabir & Parvin, 2021). Similar non-

monotonic behavior has been reported in monsoon-influenced peri-urban systems where surface conditions fluctuate despite stable land use (Jain et al., 2021).

Periods of simultaneously high urban and vegetation dominance indicate strong spatial heterogeneity across Savar, where core industrial and residential intensification coexists with persistent cropland and managed homestead vegetation (Hassan & Southworth, 2017). Because this study captures prevailing spectral-thermal surface properties relevant to LST and SUHI rather than fixed land-use classes, mixed or transitional pixels may shift dominance seasonally without implying structural land-use change (Zhou et al., 2018).

Independent ERA5-Land analysis shows no systematic anomaly in background air temperature during this period, indicating that these surface-dominance shifts are not driven by external atmospheric forcing. Together, index-based landcover dynamics' results show that urban expansion in Savar redistributes dominance among urban, vegetation, and water surfaces preferentially transforming water bodies and modulating vegetation signals without producing a uniform decline in green surfaces. This surface reconfiguration provides essential context for interpreting thermal behavior and SUHI amplification which are examined in the LST section.

Nonlinear Surface Thermal Dynamics

The evolution of land surface temperature (LST) and surface urban heat island (SUHI) intensity in Savar between 2000 and 2025 have revealed a distinctly nonlinear thermal response to urban expansion, and these strange patterns strongly modulated by monsoon-driven climate variability and peri-urban surface processes. Contrasts found in return to the linear urban warming expectations, the highest p75 LST values for both urban and vegetation-dominant surfaces occur in the early period (2000-2001). This pattern is unlikely to reflect urbanization effects and instead points to surface energy balance conditions that is associated with the pre-monsoon heat stress and reduced surface moisture availability, operating under broadly comparable atmospheric temperatures. However, ERA5-Land reanalysis confirms the absence of monotonic background air-temperature warming, which provides the insights that early extreme LST reflects surface-state sensitivity rather than long-term climatic forcing. Moreover, similar masking or reversal of urban thermal signals due to interannual climate variability has been reported in other tropical monsoon settings (Li & Bou-Zeid, 2013).

We also found the parallel cooling of urban and vegetation surfaces toward 2010-2011 further underscores the dominance of shared external forcing, likely linked to enhanced cloud cover, rainfall, or soil-moisture recharge during the warm season, i.e., analysis periods. Essentially, this synchronized LST behavior does not refute urban thermal effects but instead highlights that absolute LST trajectories alone are insufficient to isolate urban influence. Besides, the urban-vegetation contrast (Δ LST) provides a more robust indicator of surface differentiation under common atmospheric conditions. From 2015 onward, Δ LST increases sharply, despite stabilization or partial decline in absolute LST. This divergence is not accompanied by rising ERA5 air temperatures, whereas it indicates a transition toward surface-controlled thermal

differentiation(fig.....). Processes such as impervious surface consolidation, reduced evapotranspiration, and declining surface moisture increasingly dominate the urban thermal response, eventually, producing a step-change in SUHI intensity from $\sim 2^\circ\text{C}$ to $>4.5^\circ\text{C}$. Nevertheless, significant shift after 2015 has altered land-atmosphere coupling rather than regional climate warming and this findings are consistent with reported decoupling between SUHI magnitude and background climate in Dhaka and other rapidly urbanizing South Asian cities (Dewan et al., 2021) ;Uddin et al., 2022.

SUHI Amplification in a Monsoon-Driven Urbanizing Landscape

SUHI amplification in Savar is further characterized by episodic and threshold-like behavior rather than gradual scaling with urban extent. We also considered tail behavior whether the mean SUHI (p75) can be explained by p90 SUHI values. However, finding weak relationship between mean SUHI and extreme SUHI (p90; $R^2 \approx 0.15$) indicates that average conditions poorly predict tail behavior. Our findings about early periods exhibit high spatial heterogeneity ($SD \approx 5.6^\circ\text{C}$), reflecting a highly fragmented landscape with sharp contrasts between cool wet surfaces and localized hot built-up pockets. In contrast, later periods show declining variability ($SD \approx 2.1^\circ\text{C}$ by 2024-2025) alongside elevated mean SUHI. However, these findings can be attributed to the land surface dynamics in Savar from 2000 to 2025. In early periods, Savar was considerably characterized by a fragmented peri-urban landscape composed of dispersed settlements, exposed bare surfaces, agricultural fields, wetlands, and relatively limited industrial development. This heterogeneous surface configuration created strong local contrasts between cooler vegetated or water-dominated areas and warmer built-up or bare surfaces, resulting in elevated spatial variability of SUHI during the early period. In contrast, recent years show a more consolidated and spatially continuous urban structure, where built-up and industrial areas have expanded and become more interconnected. This increasing surface homogenization has reduced sharp thermal contrasts across short distances, contributing to lower spatial variability of SUHI despite higher mean SUHI intensity which are consistent with convergence patterns observed in other tropical and subtropical cities (Manoli et al., 2019); (Zhou et al., 2014); (Zhao et al., 2014).

Despite declining the overall variability, hotspot persistence analysis reveals stable thermal cores in central Savar (e.g., Savar City, Ashulia, Bipail, Amin Bazar) across multiple periods. The recurrence of these hotspots independent of interannual climatic fluctuations indicates structural controls linked to fixed land uses and urban form, such as industrial facilities, dense impervious surfaces, and high heat-storage materials. This explains the apparent contrast observed in 2010-2011, where low mean SUHI coincides with pronounced extremes, likewise, localized thermal risks persist even when citywide averages are muted. Similar separation between mean conditions and hotspot behavior has been documented where SUHI extremes are governed by land-use type and surface configuration rather than urban extent alone (Zhao et al., 2014).

Background atmospheric conditions at satellite overpass

Scene-matched ERA5-Land 2-m air temperature at Landsat overpass time shows limited inter-block variability across the six analysis periods (Fig. 12a–b: ERA5 overpass-time boxplots and mean $\pm$ SD). Median values and interquartile ranges overlap substantially, indicating broadly comparable atmospheric conditions during the April-July observation window. Mean overpass-time air temperatures remain within a narrow low-30 °C range, with no monotonic warming trend over time, though inter-block differences are present but moderate. The 2015-2016 period exhibits the lowest mean air temperature and the widest spread toward cooler values, while 2010-2011 and 2024-2025 show higher upper-tail values and larger dispersion (Fig. 12a). However, we used ERA5 analysis to understand whether the blocks represent a uniquely extreme atmospheric background relative to the others.

SUHI magnitude versus background air temperature

Comparison of mean SUHI intensity with corresponding ERA5 overpass-time air temperature reveals no systematic linear association (Fig. 13: SUHI versus ERA5 overpass-time temperature scatter). Our analysis year blocks experiencing similar atmospheric temperatures display evidently different SUHI magnitudes, and periods with relatively high SUHI do not consistently coincide with the warmest background conditions. Notably, some of the highest mean SUHI values occur under moderate atmospheric temperatures, e.g., 2000-2001 and 2015-2016 blocks, while opposite findings are also noticeable in 2005-2006 and 2010-2011 blocks. These patterns indicate that inter-block SUHI variability is not proportional to contemporaneous background air temperature.

ERA5-controlled SUHI residuals

After accounting for background atmospheric temperature using ERA5 data, a clear difference emerge between earlier and later periods (Fig. X). During 2000-2001, 2005-2006, and 2010-2011, SUHI residuals are close to zero or slightly negative, indicating that observed SUHI during these periods can largely be explained by prevailing atmospheric conditions at the time of satellite overpass. By contrast we found an opposite observation from 2015-2016 onward, where residuals become positive and increase over time, with the largest residual observed in 2024-2025. This significant findings indicate that SUHI in the later periods exceeds what would be expected from background air temperature alone, pointing to an increasing contribution from urban surface characteristics rather than atmospheric forcing.